

OpenFOAM assignment 3

Due date on Canvas

COE 347

There is absolutely no tolerance for academic misconduct. Late assignments will not be accepted, unless under exceptional circumstances at the instructor's full discretion.

Submit your assignment electronically as a PDF via Canvas by 11:59pm on the due date.

If you are submitting scanned copies of your handwritten notes, rather than a typeset document, please take the time to reduce the file to a manageable size by adjusting the resolution.

1 Objectives

In this assignment, you will simulate the compressible, two-dimensional supersonic flow in a tunnel with a forward facing step. This test case is known as “The Mach 3 wind tunnel with a step”. We shall describe the flow with the inviscid limit of the Navier-Stokes equations, i.e. the Euler equations. The configuration features supersonic and subsonic regions and you will utilize methods that are designed specifically to simulate shocks.

Because the flow features a normal shock on the leading step, you will be able to compare analytical solutions for the shock strength, stagnation pressure, and various other compressible flow parameters.

Similarly, extensive literature on this flow configuration exists, allowing for comparisons and guidance from previous work in terms of mesh resolution requirements.

You will also have the opportunity to execute the simulations “in parallel“, i.e. using multiple cores on one or more nodes.

2 Resources

You may find useful information from various resources

- Class notes on the forward facing step
- Three journal papers that show and discuss CFD solutions to this flow configuration are available on Canvas
 - Emery, An evaluation of several differencing methods for inviscid fluid flow problems, J. Comput. Phys., 2 (3) (1968), pp. 306-331
 - Woodward & Colella, The numerical simulation of two-dimensional fluid flow with strong shocks, J. Comput. Phys., 54 (1) (1984), pp. 115-173
 - Le Touze, Murrone, Guillard, Multislope MUSCL method for general unstructured meshes, J. Comput. Phys., 284 (2015), pp. 389-418
- Compressible flow book

3 Preliminaries

Pull the GitHub repository from the remote one. This can be achieved by moving into `mainclassrepo` and issuing the command `git pull`. Confirm that you now have a folder titled `OpenFOAM/current/mach-3-wind-tunnel-with-step`

4 Governing equations

The governing equations for this problem are the two-dimensional Euler equations, complemented with the equation of state for an ideal gas with $\gamma = 1.4$.

In your report:

- Write down the governing equations in vector form. Identify clearly the state vector $\mathbf{U}$, the x and y components of the flux ($\mathbf{F}$ and $\mathbf{G}$, respectively), and any additional equation that relates the variables together, e.g. the equation of state.
- Include working definitions of the local Mach number of the flow and the local sound speed. You will need those definitions when plotting quantities.
- Apply basic normal shock and isentropic flow relations to estimate the pressure at the stagnation point on the surface of the step at $(x, y) = (0.6, 0)$ for free stream conditions of $M = 3$ and $p = 1$ Pa.

5 Establish a first solution on the coarse mesh

Using the solver `rhoCentralFoam`, solve the problem for the following inlet conditions: $M = 3$, $p = 1$ Pa, $T = 1$ K, $\rho = \gamma \text{ kg/m}^3$ with $\gamma = 1.4$. Advance the solution up to $T = 4$ s.

In your report:

- Include figures (e.g. from Paraview) that provide an overview of solution. Make sure to include figures for: density ρ , x - and y - components of velocity u and v , velocity magnitude $|\mathbf{u}|$, pressure p , Mach number $M = |\mathbf{u}|/a$, sound speed a , and temperature T .

6 Converge the solution

Suppose you are asked to provide estimates for the spatial distribution of pressure on the surface of the step as well as along the top wall at time $T = 4$ s.

As you know, the pressures field depends on the spatial and temporal discretization, as well as on the method used.

For this part of the assignment, you are free to do as you think appropriate, by executing as many simulations as needed to convince yourself that you are presenting *reliable data* on pressure.

It seems reasonable to explore at least 3 levels of mesh refinement, compatibly with execution times and costs.

To report pressure along the surfaces, you may modify the file `system/singleGraph` by adding more segments that cut through the first cell next to walls as a function of either the x (horizontal surfaces) or y (vertical surfaces) coordinate.

In your report:

- Provide the details of any mesh you used during the convergence study in the form of a table. Make sure to indicate the time step used for each of those meshes.
- Produce Paraview plots that provide an overview of how spatial resolution improves the solution. Pick two variables that exhibit interesting features, i.e. pressure and density, and plot them at various levels of spatial resolution.
- In the form of $x - y$ plots, report the spatial distribution of the pressure at time $T = 4$ s on all wall surfaces, including the step vertical and horizontal faces and the top wall. If you are reporting data from different simulations to show convergence, identify them clearly.
- Comment on how the stagnation pressure at the foot of the step from the numerical solution compares to the analytical solution you found earlier for $M = 3$ and $p = 1$ Pa inlet stream.

7 Explore the performance of the parallel solver

In this last section, you will explore the performance of the parallel version of `rhoCentralFoam`. Note that the objective is to characterize the performance of the flow solver with regard to the time taken to advance over one time step.

Define the “parallel speedup” as

$$\text{Speedup} = T(1)/T(P), \quad (1)$$

where P is the number of processors used, $T(P)$ is the time taken by the solver to advance over one time step, and $T(1) = T(P = 1)$. Further, define the “parallel efficiency” η as

$$\eta = P^{-1} T(1)/T(P). \quad (2)$$

Select a “large mesh” case, say about 1 M cells or more and conduct parallel scaling tests by solving the same problem on an increasing number of processors $P = \{1, 2, 4, \dots, 64\}$ as explained in the video tutorials.

Gather and process the data from the various `log.rhoCentralFoam` files and obtain $T(P)$.

In your report:

- Produce a table of $T(P)$, speedup, and efficiency η as a function of P , with $P = \{1, 2, 4, \dots, 64\}$
- Produce a plot of the speed up vs. the number of processors P .

You can find an example of the plot of the speedup vs. processors count on the assignment page.